

Arindam Basu

professor

基本信息

姓名: Arindam Basu

所属院校: 香港城市大学

地区: China-Hong Kong

海外经历: 有

学科领域

人工智能

电子与计算机工程

生物医学工程

集成电路设计与集成系统

研究领域

深度学习与神经网络

嵌入式系统与硬件设计

低功耗电路与能量管理

生物信号处理与模式识别

脑机接口与神经工程

集成电路设计与制造

计算机体系结构

智能感知与传感器技术

机器学习理论

低功耗设计与能耗优化

个人简介

Arindam Basu is a Professor at the City University of Hong Kong, previously serving as an Associate Professor of Electrical and Electronic Engineering at Nanyang Technological University (NTU). His research focuses on Neuromorphic Engineering, Analog IC design, Computing-In-Memory systems, and Brain-machine interfaces. Basu has collaborated with numerous esteemed co-authors and has published extensively in high-impact journals, contributing significantly to the field of neuromorphic engineering and related technologies.

工作经历

City University of Hong Kong

教育经历

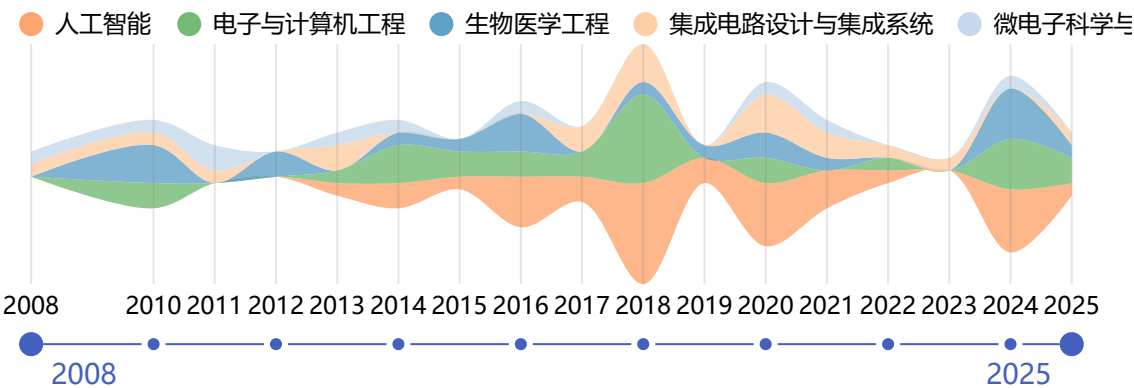
Georgia Institute of Technology - PhD

2006-01-01 - 2010-04-01

Georgia Institute of Technology - M.S

2007-01-01 - 2009-05-01

近年发表的论文



Towards neuromorphic compression based neural sensing for next-generation wireless implantable brain machine interface
NEUROMORPHIC COMPUTING AND ENGINEERING
Mohan, Vivek; Tay, Wee Peng; Basu, Arindam
被引用次数: 0 2025

Historical perspective and opportunity for computing in memory using floating-gate and resistive non-volatile computing including neuromorphic computing
NEUROMORPHIC COMPUTING AND ENGINEERING
Hasler, Jennifer; Basu, Arindam
被引用次数: 0 2025

Efficient nonlinear function approximation in analog resistive crossbars for recurrent neural networks
NATURE COMMUNICATIONS
Yang, Junyi; Mao, Ruibin; Jiang, Mingrui; Cheng, Yichuan; Sun, Pao-Sheng Vincent; Dong, Shuai; Pedretti, Giacomo; Sheng, Xia; Ignowski, Jim; Li, Haoliang; Li, Can; Basu, Arindam
被引用次数: 1 2025

Grand Challenge on Neural Decoding for Motor Control of non-Human Primates
2024 IEEE BIOMEDICAL CIRCUITS AND SYSTEMS CONFERENCE, BIOCAS 2024
Zhou, Biyan; Sun, Pao-Sheng Vincent; Yik, Jason; Frenkel, Charlotte; Reddi, Vijay Janapa; Basu, Arindam
被引用次数: 0 2024

Older and Wiser: The Marriage of Device Aging and Intellectual Property Protection of DNNs

Design Automation Conference

Ning Lin; Shaocong Wang; Yue Zhang; Yangu He; Kwunhang Wong; Arindam Basu; Dashan Shang; Xiaoming Chen; Zhongrui Wang

被引用次数: 0

2024

Halide perovskite photovoltaics for in-sensor reservoir computing

NANO ENERGY

Sharma, Divyam; Luqman, Alka; Ng, Si En; Yantara, Natalia; Xing, Xuechao; Tay, Yeow Boon; Basu, Arindam; Chattopadhyay, Anupam; Mathews, Nripan

被引用次数: 1

2024

Retinomorphic Color Perception Based on Opponent Process Enabled by Perovskite Bipolar Photodetectors

ADVANCED MATERIALS

Ng, Si En; Yantara, Natalia; Tu, Ngo Anh; Erdenebileg, Enkhtur; Li, Patrick Wen Feng; Sharma, Divyam; Lam, Yeng Ming; Mhaisalkar, Subodh; Basu, Arindam; Chattopadhyay, Anupam; Mathews, Nripan

被引用次数: 0

2024

Hybrid Event-Frame Neural Spike Detector for Neuromorphic Implantable BMI

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Mohan, Vivek; Tay, Wee Peng; Basu, Arindam

被引用次数: 0

2024

A Low-Power Spike Detector Using In-Memory Computing for Event-based Neural Frontend

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Ke, Ye; Basu, Arindam

被引用次数: 0

2024

Exploiting deep learning accelerators for neuromorphic workloads

NEUROMORPHIC COMPUTING AND ENGINEERING

Sun, Pao-Sheng Vincent; Titterton, Alexander; Gopiani, Anjee; Santos, Tim; Basu, Arindam; Lu, Wei D.; Eshraghian, Jason K.

被引用次数: 2

2024

High-performance one-dimensional halide perovskite crossbar memristors and synapses for neuromorphic computing

MATERIALS HORIZONS

Vishwanath, Sujaya Kumar; Febriansyah, Benny; Ng, Si En; Das, Tisita; Acharya, Jyotibdha; John, Rohit Abraham; Sharma, Divyam; Dananjaya, Putu Andhita; Jagadeeswararao, Metikoti; Tiwari, Naveen; Kulkarni, Mohit Ramesh Chandra; Lew, Wen Siang; Chakraborty, Sudip; Basu, Arindam; Mathews, Nripan

被引用次数: 10

2024

A Hybrid Neuromorphic Object Tracking and Classification Framework for Real-Time Systems

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Ussa, Andres; Rajen, Chockalingam Senthil; Pulluri, Tarun; Singla, Deepak; Acharya, Jyotibdha; Chuanrong, Gideon Fu; Basu, Arindam; Ramesh, Bharath

被引用次数: 3

2024

HAST: A Hardware-Efficient Spatio-Temporal Correlation Near-Sensor Noise Filter for Dynamic Vision Sensors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Gopalakrishnan, Pradeep Kumar; Chang, Chip-Hong; Basu, Arindam

被引用次数: 0

2024

A 389 TOPS/W, Always ON Region Proposal Integrated Circuit Using In-Memory Computing in 65 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bose, Sumon Kumar; Basu, Arindam

被引用次数: 5

2023

EBBINNOT: A Hardware-Efficient Hybrid Event-Frame Tracker for Stationary Dynamic Vision Sensors

IEEE INTERNET OF THINGS JOURNAL

Mohan, Vivek; Singla, Deepak; Pulluri, Tarun; Ussa, Andres; Gopalakrishnan, Pradeep Kumar; Sun, Pao-Sheng; Ramesh, Bharath; Basu, Arindam

被引用次数: 10

2022

A 51.3-TOPS/W, 134.4-GOPS In-Memory Binary Image Filtering in 65-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bose, Sumon Kumar; Singla, Deepak; Basu, Arindam

被引用次数: 9

2022

Dendritic Computing: Branching Deeper into Machine Learning

NEUROSCIENCE

Acharya, Jyotibdha; Basu, Arindam; Legenstein, Robert; Limbacher, Thomas; Poirazi, Panayiota; Wu, Xundong

被引用次数: 19

2022

DeepFreeze: Cold Boot Attacks and High Fidelity Model Recovery on Commercial EdgeML Device

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Won, Yoo-Seung; Chatterjee, Soham; Jap, Dirmanto; Basu, Arindam; Bhasin, Shivam

被引用次数: 4

2021

Power-efficient Spike Sorting Scheme Using Analog Spiking Neural Network Classifier

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Mukhopadhyay, Anand Kumar; Sharma, Atul; Chakrabarti, Indrajit; Basu, Arindam; Sharad, Mrigank

被引用次数: 7

2021

A 0.11-0.38 pJ/cycle Differential Ring Oscillator in 65 nm CMOS for Robust Neurocomputing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhang, Xueyong; Acharya, Jyotibdha; Basu, Arindam

被引用次数: 8

2021

Introducing Recurrence in Strong PUFs for Enhanced Machine Learning Attack Resistance

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Shah, Nimesh; Chatterjee, Durba; Sapui, Brojogopal; Mukhopadhyay, Debdeep; Basu, Arindam

被引用次数: 18

2021

Halide perovskite memristors as flexible and reconfigurable physical unclonable functions

NATURE COMMUNICATIONS

John, Rohit Abraham; Shah, Nimesh; Vishwanath, Sujaya Kumar; Ng, Si En; Febriansyah, Benny; Jagadeeswararao, Metikoti; Chang, Chip-Hong; Basu, Arindam; Mathews, Nripan

被引用次数: 126

2021

Diffusive and Drift Halide Perovskite Memristive Barristors as Nociceptive and Synaptic Emulators for Neuromorphic Computing

ADVANCED MATERIALS

John, Rohit Abraham; Yantara, Natalia; Ng, Si En; Patdillah, Muhammad Iszaki Bin; Kulkarni, Mohit Rameshchandra; Jamaludin, Nur Fadilah; Basu, Joydeep; Ankit; Mhaisalkar, Subodh G.; Basu, Arindam; Mathews, Nripan

被引用次数: 224

2021

A 389TOPS/W, 1262fps at 1Meps Region Proposal Integrated Circuit for Neuromorphic Vision Sensors in 65nm CMOS

IEEE ASIAN SOLID-STATE CIRCUITS CONFERENCE (A-SSCC 2021)

Bose, Sumon Kumar; Basu, Arindam

被引用次数: 1

2021

A 125 μ W Readout Circuit in 65 nm CMOS With Successive Approximation-Based Thresholding for Domain Wall Magnet-Based Random Number Generator

IEEE SENSORS JOURNAL

Narasimman, Govind; Basu, Joydeep; Sethi, Pankaj; Krishnia, Sachin; Yi, Chen; Siang, Lew Wen; Basu, Arindam

被引用次数: 4

2020

Self healable neuromorphic memtransistor elements for decentralized sensory signal processing in robotics

NATURE COMMUNICATIONS

John, Rohit Abraham; Tiwari, Naveen; Bin Patdillah, Muhammad Iszaki; Kulkarni, Mohit Rameshchandra; Tiwari, Nidhi; Basu, Joydeep; Bose, Sumon Kumar; Ankit; Yu, Chan Jun; Nirmal, Amoolya; Vishwanath, Sujaya Kumar; Bartolozzi, Chiara; Basu, Arindam; Mathews, Nripan

被引用次数: 80

2020

HFNet: A CNN Architecture Co-designed for Neuromorphic Hardware With a Crossbar Array of Synapses

FRONTIERS IN NEUROSCIENCE

Gopalakrishnan, Roshan; Chua, Yansong; Sun, Pengfei; Sreejith Kumar, Ashish Jith; Basu, Arindam

被引用次数: 12

2020

Sparse Ensemble Machine Learning to Improve Robustness of Long-Term Decoding in iBMIs

IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING

Shaikh, Shoeb; So, Rosa; Sibindi, Tafadzwa; Libedinsky, Camilo; Basu, Arindam

被引用次数: 6

2020

Optogenetics inspired transition metal dichalcogenide neuristors for in-memory deep recurrent neural networks

NATURE COMMUNICATIONS

John, Rohit Abraham; Acharya, Jyotibdha; Zhu, Chao; Surendran, Abhijith; Bose, Sumon Kumar; Chaturvedi, Apoorva; Tiwari, Nidhi; Gao, Yang; He, Yongmin; Zhang, Keke K.; Xu, Manzhong; Leong, Wei Lin; Liu, Zheng; Basu, Arindam; Mathews, Nripan

被引用次数: 43

2020

Deep Neural Network for Respiratory Sound Classification in Wearable Devices Enabled by Patient Specific Model Tuning

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Acharya, Jyotibdha; Basu, Arindam

被引用次数: 126

2020

ADIC: Anomaly Detection Integrated Circuit in 65-nm CMOS Utilizing Approximate Computing

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kar, Bapi; Gopalakrishnan, Pradeep Kumar; Bose, Sumon Kumar; Roy, Mohendra; Basu, Arindam

被引用次数: 6

2020

CRAM: Collocated SRAM and DRAM With In-Memory Computing-Based Denoising and Filling for Neuromorphic Vision Sensors in 65 nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Xueyong; Mohan, Vivek; Basu, Arindam

被引用次数: 5

2020

ADEPOS: A Novel Approximate Computing Framework for Anomaly Detection Systems and its Implementation in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bose, Sumon Kumar; Kar, Bapi; Roy, Mohendra; Gopalakrishnan, Pradeep Kumar; Zhang, Lei; Patil, Aakash; Basu, Arindam

被引用次数: 18

2020

A 2.86-TOPS/W Current Mirror Cross-Bar-Based Machine-Learning and Physical Unclonable Function Engine For Internet-of-Things Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Yi; Wang, Zheng; Patil, Aakash; Basu, Arindam

被引用次数: 22

2019

Guest Editorial: Special Issue on Selected Papers From IEEE ISICAS 2019

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Kiani, M.; Basu, A.; Lee, S. -Y.

被引用次数: 0

2019

Towards Intelligent Intracortical BMI Low-Power Neuromorphic Decoders That Outperform Kalman Filters

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Shaikh, Shoeb; So, Rosa; Sibindi, Tafadzwa; Libedinsky, Camilo; Basu, Arindam

被引用次数: 16

2019

Fifty-eighth Joint Technological Conference of ATIRA, BTRA, SITRA and NITRA

INDIAN JOURNAL OF FIBRE & TEXTILE RESEARCH

Gaur, R. K.; Parmar, M. S.; Basu, A.

被引用次数: 0

2019

Synergistic Gating of Electro-Iono-Photoactive 2D Chalcogenide Neuristors: Coexistence of Hebbian and Homeostatic Synaptic Metaplasticity

ADVANCED MATERIALS

John, Rohit Abraham; Liu, Fucui; Nguyen Anh Chien; Kulkarni, Mohit R.; Zhu, Chao; Fu, Qundong; Basu, Arindam; Liu, Zheng; Mathews, Nripan

被引用次数: 289

2018

A Comparison of Low-Complexity Real-Time Feature Extraction for Neuromorphic Speech Recognition

FRONTIERS IN NEUROSCIENCE

Acharya, Jyotibdha; Patil, Aakash; Li, Xiaoya; Chen, Yi; Liu, Shih-Chii; Basu, Arindam

被引用次数: 14

2018

A Noise Filtering Algorithm for Event-Based Asynchronous Change Detection Image Sensors on TrueNorth and Its Implementation on TrueNorth

FRONTIERS IN NEUROSCIENCE

Padala, Vandana; Basu, Arindam; Orchard, Garrick

被引用次数: 54

2018

Spiking Neural Classifier with Lumped Dendritic Nonlinearity and Binary Synapses: A Current Mode VLSI Implementation and Analysis

NEURAL COMPUTATION

Bhaduri, Aritra; Banerjee, Amitava; Roy, Subhrajit; Kar, Sougata; Basu, Arindam

被引用次数: 17

2018

An Extreme Learning Machine-Based Neuromorphic Tactile Sensing System for Texture Recognition

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Rasouli, Mahdi; Chen, Yi; Basu, Arindam; Kukreja, Sunil L.; Thakor, Nitish V.

被引用次数: 63

2018

Domain Wall Motion-Based Dual-Threshold Activation Unit for Low-Power Classification of Non-Linearly Separable Functions

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Deb, Suman; Vatwani, Tarun; Chattopadhyay, Anupam; Basu, Arindam; Fong, Xuanyao

被引用次数: 1

2018

Ionotronic Halide Perovskite Drift-Diffusive Synapses for Low-Power Neuromorphic Computation

ADVANCED MATERIALS

John, Rohit Abraham; Yantara, Natalia; Ng, Yan Fong; Narasimman, Govind; Mosconi, Edoardo; Meggiolaro, Daniele; Kulkarni, Mohit R.; Gopalakrishnan, Pradeep Kumar; Nguyen, Chien A.; De Angelis, Filippo; Mhaisalkar, Subodh G.; Basu, Arindam; Mathews, Nripan

被引用次数: 171

2018

Current Mirror Array: A Novel Circuit Topology for Combining Physical Unclonable Function and Machine Learning

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Zheng; Chen, Yi; Patil, Aakash; Jayabalan, Jayasanker; Zhang, Xueyong; Chang, Chip-Hong; Basu, Arindam

被引用次数: 32

2018

Guest Editorial Low-Power, Adaptive Neuromorphic Systems: Devices, Circuit, Architectures and Algorithms

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Basu, Arindam; Chang, Meng-Fan; Chicca, Elisabetta; Karnik, Tanay; Li, Hai; Seo, Jae-Sun

被引用次数: 0

2018

Low-Power, Adaptive Neuromorphic Systems: Recent Progress and Future Directions

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Basu, Arindam; Acharya, Jyotibdha; Karnik, Tanay; Liu, Huichu; Li, Hai; Seo, Jae-Sun; Song, Chang

被引用次数: 78

2018

Ultralow Power Dual-Gated Subthreshold Oxide Neuristors: An Enabler for Higher Order Neuronal Temporal Correlations

ACS NANO

John, Rohit Abraham; Tiwari, Nidhi; Chen Yaoyi; Ankit; Tiwari, Naveen; Kulkarni, Mohit; Nirmal, Arnoolya; Anh Chien Nguyen; Basu, Arindam; Mathews, Nripan

被引用次数: 74

2018

Special Issue on Neuromorphic Computing and Cognitive Systems

IEEE TRANSACTIONS ON COGNITIVE AND DEVELOPMENTAL SYSTEMS

Tang, Huajin; Huang, Tiejun; Krichmar, Jeffrey L.; Orchard, Garrick; Basu, Arindam

被引用次数: 2

2018

Hardware architecture for large parallel array of Random Feature Extractors applied to image recognition

NEUROCOMPUTING

Patil, Aakash; Shen, Shanlan; Yao, Enyi; Basu, Arindam

被引用次数: 17

2017

VLSI Extreme Learning Machine: A Design Space Exploration

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yao, Enyi; Basu, Arindam

被引用次数: 35

2017

An Online Unsupervised Structural Plasticity Algorithm for Spiking Neural Networks

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Roy, Subhrajit; Basu, Arindam

被引用次数: 30

2017

Triplet Spike Time-Dependent Plasticity in a Floating-Gate Synapse

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Gopalakrishnan, Roshan; Basu, Arindam

被引用次数: 17

2017

A 0.7 V, 40 nW Compact, Current-Mode Neural Spike Detector in 65 nm CMOS

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Yao, Enyi; Chen, Yi; Basu, Arindam

被引用次数: 26

2016

Multiclass Classification by Adaptive Network of Dendritic Neurons with Binary Synapses Using Structural Plasticity

FRONTIERS IN NEUROSCIENCE

Hussain, Shaista; Basu, Arindam

被引用次数: 8

2016

An Online Structural Plasticity Rule for Generating Better Reservoirs

NEURAL COMPUTATION

Roy, Subhrajit; Basu, Arindam

被引用次数: 14

2016

Learning Spike Time Codes Through Morphological Learning With Binary Synapses

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Roy, Subhrajit; San, Phyo Phyo; Hussain, Shaista; Wei, Lee Wang; Basu, Arindam

被引用次数: 13

2016

A 128-Channel Extreme Learning Machine-Based Neural Decoder for Brain Machine Interfaces

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Chen, Yi; Yao, Enyi; Basu, Arindam

被引用次数: 86

2016

Guest Editorial-Special Issue on Selected Papers From IEEE BioCAS 2015

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Ostadabbas, Sarah; Demarchi, Danilo; Basu, Arindam

被引用次数: 0

2016

Hardware-Amenable Structural Learning for Spike-Based Pattern Classification Using a Simple Model of Active Dendrites

NEURAL COMPUTATION

Hussain, Shaista; Liu, Shih-Chii; Basu, Arindam

被引用次数: 18

2015

On the Non-STDP Behavior and Its Remedy in a Floating-Gate Synapse

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Gopalakrishnan, Roshan; Basu, Arindam

被引用次数: 5

2015

A Digitally Assisted, Signal Folding Neural Recording Amplifier

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Chen, Yi; Basu, Arindam; Liu, Lei; Zou, Xiaodan; Rajkumar, Ramamoorthy; Dawe, Gavin Stewart; Je, Minkyu

被引用次数: 35

2014

Liquid State Machine With Dendritically Enhanced Readout for Low-Power, Neuromorphic VLSI Implementations

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Roy, Subhrajit; Banerjee, Amitava; Basu, Arindam

被引用次数: 43

2014

A 0.7 V low-power fully programmable Gaussian function generator for brain-inspired Gaussian correlation associative memory

NEUROCOMPUTING

Li, Fei; Chang, Chip-Hong; Basu, Arindam; Siek, Liter

被引用次数: 7

2014

Delay learning architectures for memory and classification

NEUROCOMPUTING

Hussain, Shaista; Basu, Arindam; Wang, Runchun Mark; Hamilton, Tara Julia

被引用次数: 15

2014

Speech Processing on a Reconfigurable Analog Platform

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ramakrishnan, Shubha; Basu, Arindam; Chiu, Leung Kin; Hasler, Jennifer; Anderson, David; Brink, Stephen

被引用次数: 9

2014

A Learning-Enabled Neuron Array IC Based Upon Transistor Channel Models of Biological Phenomena

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Brink, Stephen; Nease, Stephen; Hasler, Paul; Ramakrishnan, Shubha; Wunderlich, Richard; Basu, Arindam; Degnan, Brian

被引用次数: 94

2013

Models for Characterizing Noise Based PCMOS Circuits

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Singh, Anshul; Basu, Arindam; Ling, Keck-Voon; Mooney, Vincent J., III

被引用次数: 1

2013

Silicon spiking neurons for hardware implementation of extreme learning machines

NEUROCOMPUTING

Basu, Arindam; Shuo, Sun; Zhou, Hongming; Lim, Meng Hiot; Huang, Guang-Bin

被引用次数: 62

2013

Small-Signal Neural Models and Their Applications

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Basu, Arindam

被引用次数: 8

2012

Dynamical Systems Guided Design and Analysis of Silicon Oscillators for Central Pattern Generators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Li, Fei; Basu, Arindam; Chang, Chip-Hong; Cohen, Avis H.

被引用次数: 8

2012

Low-Power Discrete Fourier Transform for OFDM: A Programmable Analog Approach

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Suh, Sangwook; Basu, Arindam; Schlottmann, Craig; Hasler, Paul E.; Barry, John R.

被引用次数: 39

2011

A Fully Integrated Architecture for Fast and Accurate Programming of Floating Gates Over Six Decades of Current

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Basu, Arindam; Hasler, Paul E.

被引用次数: 31

2011

Dynamics and Bifurcations in a Silicon Neuron

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Basu, Arindam; Petre, Csaba; Hasler, Paul E.

被引用次数: 9

2010

Neural Dynamics in Reconfigurable Silicon

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Basu, Arindam; Ramakrishnan, Shubha; Petre, Csaba; Koziol, Scott; Brink, Stephen; Hasler, Paul E.

被引用次数: 75

2010

A Floating-Gate-Based Field-Programmable Analog Array

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Basu, Arindam; Brink, Stephen; Schlottmann, Craig; Ramakrishnan, Shubha; Petre, Csaba; Koziol, Scott; Baskaya, Faik; Twigg, Christopher M.; Hasler, Paul

被引用次数: 113

2010

Nullcline-Based Design of a Silicon Neuron

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Basu, Arindam; Hasler, Paul E.

被引用次数: 60

2010

Magnetic performance of the main superconducting magnets for the LHC

IEEE TRANSACTIONS ON APPLIED SUPERCONDUCTIVITY

Sanfilippo, S.; Sammut, N.; Bottura, L.; Di-Castro, A.; Basu, A.; Koutchouk, J. -P.; Todesco, E.; Hagen, P.; Catalan-Lasheras, N.; Venturini-Delsolaro, W.; Giloux, C.; Wolf, R.

被引用次数: 3

2008

A charge-based low-power high-SNR capacitive sensing interface circuit

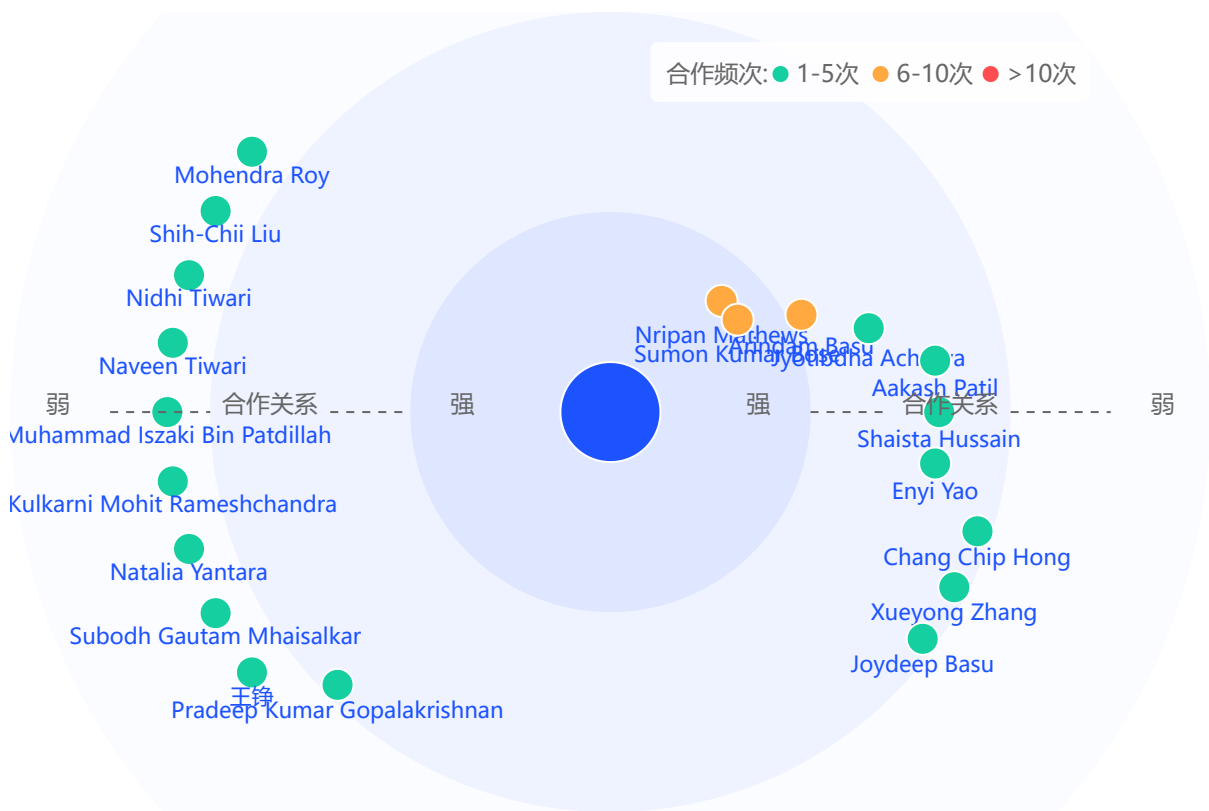
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Peng, Sheng-Yu; Qureshi, Muhammad S.; Hasler, Paul E.; Basu, Arindam; Degertekin, F. Levent

被引用次数: 61

2008

学术合作



Nripan Mathews

南洋理工大学

合作次数: 7

Sumon Kumar Bose

南洋理工大学

合作次数: 7

Arindam Basu

佐治亚理工学院

合作次数: 6

Jyotibdha Acharya

南洋理工大学

合作次数: 5

Aakash Patil

南洋理工大学

合作次数: 4

还有 48 个合作者